Lesson 9: Texture Mapping

Lesson 9: Texture Mapping 1
9.1 Texture Sources 2
9.2 Mapping from Surfaces into Texture Space 3
9.2.1 Basic Concepts 3
9.3 Textures and Phong Reflectance 3
9.4 Aliasing 4
9.5 Texturing in OpenGL 4
Lesson 9 Discussion Session 5

9.1 Texture Sources

Texture is the appearance and feel of a surface The term also refers to an image used to define the characteristics of a surface, which is a multidimensional image mapped to a multidimensional space.

A 2D texture in OpenGL is a 2D array of colors, and each element in the array is called a 'texel. We tell OpenGL how to take texel colors from the array during rendering, and use them to influence the pixel colors.

Texture Definition

From a programming point of view, texture can be defined using two approaches;

- ✓ Hard-code the texel values into the texture array.
- ✓ Use an existing image (a JPG, PNG, GIF, BMP, etc.), and read its pixels into the texture array.

A texture unit, also called a texture mapping unit (TMU) or a texture processing unit (TPU), is a hardware component in a GPU that does sampling?
Sampling is the process of computing a color from an image texture and texture coordinates. Mapping a texture image to a surface is a fairly complex operation, since it requires more than just returning the color of the texel that
contains some given texture coordinates.

The texture unit is the processor; the texture object holds the data that is processed.

3D texture

A texture map that is a function of three variables. This is also called a "solid texture", because the texture map is a volume. Solid textures have been used in diffuse color texture mapping to simulate things like marble and wood grain. Sometimes specifying the color in a volume is simpler than specifying it on the surface of an object, especially if the object has a complex shape.

Texture Mapping

Definitions

A mapping from any function onto a surface in three dimension.
Most general: The mapping of any image into multidimensional space.
The process where some parameter of the apparent color computation is replaced by an external function, called the texture map.
A common example is texture mapping an object's diffuse color from an image. This gives the appearance of the image painted onto the object.

Texture Mapping Characteristics

- ✓ Too much detail to model geometrically, like grass, etc.
- ✓ Pattern is repeated (periodic)

Texture Mapping Techniques

- ✓ 2D texture mapping: paint 2D pattern onto the surface
- ✓ Environmental (reflection) mapping

- ✓ Bump mapping: perturb surface normals to fool shading algorithms
- ✓ Procedural texture mapping

Steps for Texture Mapping:

- ✓ Define texture
- ✓ Specify mapping from texture to surface
- ✓ Lookup texture values during scan conversion

Texture mapping, diffuse color is a form of texture mapping where the texture defines the object's diffuse color. This gives the
appearance of pasting the texture image onto the object.

Texture mapping is in fact a 2D projective transformation

- ✓ Texture coordinate system: (u,v) to
- ✓ Screen coordinate system (t,s)

9.2 Mapping from Surfaces into Texture Space

9.2.1 Basic Concepts

Mapping from screen to texture coordinates.

Once a point on the surface of the model has been mapped to a value in the texture,

change its RGB value (or something else!) accordingly. This is called parametric texture mapping. A single point in the texture is called a texel.

The first known use of texture in graphics was the modulation of surface color values,

(diffuse coefficients) by Catmull in 1974.

NB A texture does not have to indicate color!

Object Space to Screen

- ✓ The texture coordinates are known in the object space.
- ✓ Needs to be interpolated in the screen space.

9.3 Textures and Phong Reflectance

The Phong model calculates the amount of reflected radiance at a surface point according to the lightning, viewing, and surface normal directions at that point.

Characteristics of Phong Reflectance

Modeling specular reflection as an exponential function.

Reflectance models are mathematical functions of lighting, viewing and surface normal direction.

Phong Illumination Modell

- ✓ Calculate color for arbitrary point on surface
- ✓ Basic inputs are material properties and l, n, v:

Summary off Phong Modell

- ✓ Light components for each color:

– Ambient (L_a), diffuse (L_d), specular (L_s)

- ✓ Material coefficients for each color:

– Ambient (k_a), diffuse (k_d), specular (k_s)

- ✓ Distance q for surface point from light source

9.4 Aliasing

Aliasing

The phenomenon that makes smooth lines and edges appear stair stepped or jagged. Aliasing is also called the "jaggies".

Temporal aliasing

Aliasing in time by animation frame, instead of spatially by pixel. The visual effect of temporal aliasing has been called "strobing". Moving objects appear to jump thru a sequence of frozen steps, instead of in smooth paths.

9.5 Texturing in OpenGL

There are 6 steps to using textures in OpenGL:

1.Enable texturing

2.Make a 2D array of colors for the texture

3.Set up parameters for how the texture wraps.

4.Say how the texture combines with other material properties

5.Create the texture

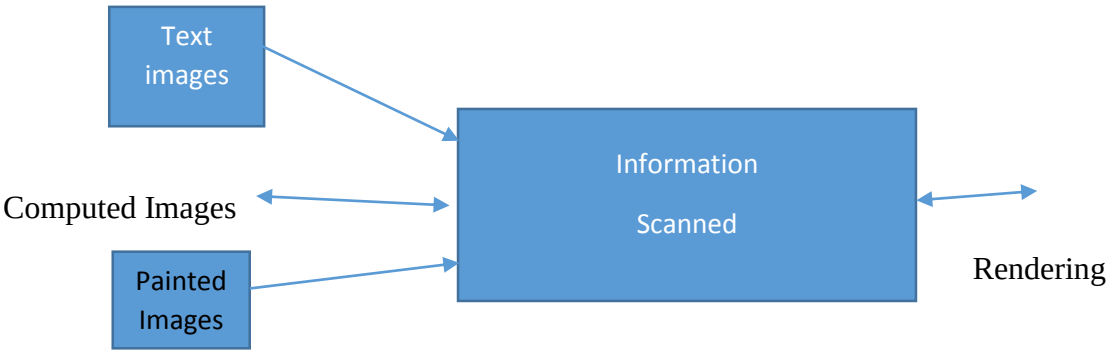
6.Give texture coordinates to go with your geometry. The code in the following explanations is taken from the TexSimpleFloat.java program.

To enable texturing:

gl.glEnable(GL2.GL_TEXTURE_2D);

Note that OpenGL also supports 3D textures. 3D textures are interesting and useful, but the textures are harder to create and we aren't going to use them this term.

The Graphics Process: Texturing



Lesson 9 Discussion Session

Discuss the following algorithms

1. DDA Algorithm.
2. Bresenham's Line Algorithm